Computational and Informatics Resources and Tools for Glycoscience Research

Project Number
5U01GM125267-04

Former Number
1U01OD024850-01

Contact PI/Project Leader
TIEMEYER, MICHAEL
Other PIs

Awardee Organization
UNIVERSITY OF GEORGIA

Description

Abstract Text

PROJECT SUMMARY/ABSTRACT Although ongoing technical advances are accelerating the pace and sophistication of data acquisition in glycoscience, the transformation of these data to glycobiology knowledge, insight, and understanding is slowed by the limited number of tools that facilitate their integration with biological knowledge from genetics, proteomics, pathology, and other disciplines. Our grant application describes the development of an integrated, extendable, and cross-disciplinary resource providing tools and data to address specific scientific questions that can currently be answered only by extensive literature-based research and manual collection of data from disparate databases and websites. Using insight gained during our planning grant activities, including a workshop focused on evaluating existing resources and community needs, we propose to develop a broadly relevant and sustainable glycoinformatics resource to connect glycoscience with the explosion of data that is revolutionizing biology. We identified critical gaps that need to be filled and challenges that must be overcome to create an enduring and sustainable glycoinformatics resource that goes beyond mapping glycan data to genes and proteins to identify and integrate diverse multidisciplinary knowledge from EMBL-EBI, NCBI, UniProt, UniCarbKB, CAZy, Gene Ontology and other sources. To maximize synergy among these resources, we propose a new glycan array data repository and enhanced ontologies to facilitate integration of glycan and glycoconjugate expression and interaction data with other information. Evaluating these data in the context of knowledge about genetic mutations, gene expression, protein function and other phenomena will provide new opportunities for systems-level understanding of the roles of glycosylation in disease and development. This comprehensive data integration framework will provide unprecedented support for complex queries spanning diverse data types relevant to glycobiology. Technical advances required to implement this framework include evidence tagging of data, ontology and standards development, and new interfaces that enable data mining, sharing, and dissemination. Community engagement, especially with scientists who do not specialize in glycobiology, will be emphasized to maximize the relevance of our resource. We will develop a portal to make all this information publicly available in standard formats supported by NCBI and EMBL-EBI and in new formats we develop, promoting sharing of data and their ultimate integration into these widely used informatics resources.

Public Health Relevance Statement

NARRATIVE Integration and dissemination of carbohydrate and glycoconjugate related data and analysis tools for knowledge discovery and wider usage by the scientific community.

NIH Spending Category

Data Science | Genetics | Networking and Information Technology R&D (NITRD)

Project Terms

Address | Anabolism | Applications Grants | Big Data | Binding | Biological | Biology | Biophysics | Carbohydrates | Code | Communities | Complex | Computer software | Coupled | DNA Sequence Alteration | Data | Data Analyses | Data Collection | Data Set | Databases | Development | Discipline | Disease | Documentation | Educational workshop | Enzymes | Explosion | Funding | Gene Expression | Gene Proteins | Genes | Genetic | Genomics | Glycobiology | Glycoconjugates | Grant | Guidelines | Infrastructure

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Intuition Knowledge Knowledge Discovery Link Lipids Literature Manuals Metadata Molecular Mutation Names Ontology Pathology Pathway interactions Polysaccharides Property Proteomics Protocols documentation Reaction Research Resource Informatics Resources Retrieval Role Scientist Seeds Source Source Code Structure System Training Visualization Visualization software base data acquisition data integration data mining data sharing data warehouse diverse data experience glycosylation hackathon insight multidisciplinary protein function repository synergism tool web portal web services web site

Details

Contact PI/ Project Leader

Name
TIEMEYER, MICHAEL
Title
PROFESSOR
Contact
mtiemeyer@ccrc.uga.edu

Other PIs

Name
MAZUMDER, RAJA

Program Official

Name
MELILLO, AMANDA A
Contact
melilloaa@mail.nih.gov

Organization

Name
UNIVERSITY OF GEORGIA
City
ATHENS
Country
UNITED STATES (US)
Department Type
NONE
Organization Type
ORGANIZED RESEARCH UNITS
State Code
GA
Congressional District
10

Other Information

Opportunity Number
RFA-RM-16-021
Study Section
Special Emphasis Panel[ZRG1-IMST-U(50)R]
Fiscal Year
2020
Award Notice Date
02-June-2020
Administering Institutes or Centers
National Institute of General Medical Sciences
CFDA Code
310
DUNS Number
004315578
UEI
NMJHD63STRC5
Project Start Date **01-September-2017**
Project End Date **31-May-2022**
Budget Start Date **01-June-2020**
Budget End Date **31-May-2021**

Project Funding Information for 2020

Total Funding
$1,892,436
Direct Costs
$1,627,031
Indirect Costs
$265,405

Year	Funding IC	FY Total Cost by IC
2020	NIH Office of the Director	$1,892,436

NIH Categorical Spending

Click here for more information on NIH Categorical Spending

Funding IC	FY Total Cost by IC	NIH Spending Category
COMMON FUND, OFFICE OF THE DIRECTOR	$1,892,436	Data Science; Genetics; Networking and Information Technology R&D (NITRD)

Sub Projects

No Sub Projects information available for 5U01GM125267-04

Publications

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Journal (Link to PubMed abstract)	Authors	Publication Year	Similar Publications	CitedBy	iCit
The Glycan Structure Dictionary-a dictionary describing commonly used glycan structure terms.					
Glycobiology 2023 Jun 03; 33 (5) 354-357	Vora, Jeet; Navelkar, Rahi; Vijay-Shanker, K; Edwards, Nathan; Martinez, Karina; Ding, Xiying; Wang, Tianyi; Su, Peng; Ross, Karen; Lisacek, Frederique; Hayes, Catherine; Kahsay, Robel; Ranzinger, Rene; Tiemeyer, Michael; Mazumder, Raja View Less	2023			
CarbArrayART: a new software tool for carbohydrate microarray data storage, processing, presentation, and reporting.					
Glycobiology 2022 Jun 13; 32 (7) 552-555	Akune, Yukie; Arpinar, Sena; Silva, Lisete M; Palma, Angelina S; Tajadura-Ortega, Virginia; Aoki-Kinoshita, Kiyoko F; Ranzinger, René; Liu, Yan; Feizi, Ten View Less	2022			
The international glycan repository GlyTouCan version 3.0.					
Nucleic acids research 2021 01 08; 49 (D1) D1529-D1533	Fujita, Akihiro; Aoki, Nobuyuki P; Shinmachi, Daisuke; Matsubara, Masaaki; Tsuchiya, Shinichiro; Shiota, Masaaki; Ono, Tamiko; Yamada, Issaku; Aoki-Kinoshita, Kiyoko F View Less	2021			iCite 4.54
SugarDrawer: A Web-Based Database Search Tool with Editing Glycan Structures.					
Molecules (Basel, Switzerland) 2021 Nov 25; 26 (23)	Tsuchiya, Shinichiro; Matsubara, Masaaki; Aoki-Kinoshita, Kiyoko F; Yamada, Issaku	2021			iCite 0.34
The glycoconjugate ontology (GlycoCoO) for standardizing the annotation of glycoconjugate data and its application.					
Glycobiology 2021 08 07; 31 (7) 741-750	Yamada, Issaku; Campbell, Matthew P; Edwards, Nathan; Castro, Leyla Jael; Lisacek, Frederique; Mariethoz, Julien; Ono, Tamiko; Ranzinger, Rene; Shinmachi, Daisuke; Aoki-Kinoshita, Kiyoko F View Less	2021			iCite 0.74
Bioinformatics tools developed to support BioCompute Objects.					
Database : the journal of biological databases and curation 2021 03 30; 2021	Patel, Janisha A; Dean, Dennis A; King, Charles Hadley; Xiao, Nan; Koc, Soner; Minina, Ekaterina; Golikov, Anton; Brooks, Phillip; Kahsay, Robel; Navelkar, Rahi; Ray, Manisha; Roberson, Dave; Armstrong, Chris; Mazumder, Raja; Keeney, Jonathon View Less	2021			iCite 0.74
The human O-GlcNAcome database and meta-analysis.					
Scientific data 2021 01 21; 8 (1) 25	Wulff-Fuentes, Eugenia; Berendt, Rex R; Massman, Logan; Danner, Laura; Malard, Florian; Vora, Jeet; Kahsay, Robel; Olivier-Van Stichelen, Stephanie	2021			iCite 11.38

Journal (Link to PubMed abstract)	Authors	Publication Year	Similar Publications	CitedBy	iCite RCR
Mucin O-glycans suppress quorum-sensing pathways and genetic transformation in Streptococcus mutans.					
Nature microbiology 2021 May; 6 (5) 574-583	Werlang, Caroline A; Chen, Wesley G; Aoki, Kazuhiro; Wheeler, Kelsey M; Tymm, Carly; Mileti, Cassidy J; Burgos, Ana C; Kim, Kris; Tiemeyer, Michael; Ribbeck, Katharina View Less	2021			iCite 2.76
Global mapping of glycosylation pathways in human-derived cells.					
Developmental cell 2021 04 19; 56 (8) 1195-1209.e7	Huang, Yi-Fan; Aoki, Kazuhiro; Akase, Sachiko; Ishihara, Mayumi; Liu, Yi-Shi; Yang, Ganglong; Kizuka, Yasuhiko; Mizumoto, Shuji; Tiemeyer, Michael; Gao, Xiao-Dong; Aoki-Kinoshita, Kiyoko F; Fujita, Morihisa View Less	2021			iCite 4.60
Enhancing the interoperability of glycan data flow between ChEBI, PubChem and GlyGen.					
Glycobiology 2021 12 18; 31 (11) 1510-1519	Navelkar, Rahi; Owen, Gareth; Mutherkrishnan, Venkatesh; Thiessen, Paul; Cheng, Tiejun; Bolton, Evan; Edwards, Nathan; Tiemeyer, Michael; Campbell, Matthew P; Martin, Maria; Vora, Jeet; Kahsay, Robel; Mazumder, Raja View Less	2021			iCite 0.50
The GlyCosmos Portal: a unified and comprehensive web resource for the glycosciences.					
Nature methods 2020 07; 17 (7) 649-650	Yamada, Issaku; Shiota, Masaaki; Shinmachi, Daisuke; Ono, Tamiko; Tsuchiya, Shinichiro; Hosoda, Masae; Fujita, Akihiro; Aoki, Nobuyuki P; Watanabe, Yu; Fujita, Noriaki; Angata, Kiyohiko; Kaji, Hiroyuki; Narimatsu, Hisashi; Okuda, Shujiro; Aoki-Kinoshita, Kiyoko F View Less	2020			iCite 4.91
GlyGen: Computational and Informatics Resources for Glycoscience.					
Glycobiology 2020 01 28; 30 (2) 72-73	York, William S; Mazumder, Raja; Ranzinger, Rene; Edwards, Nathan; Kahsay, Robel; Aoki-Kinoshita, Kiyoko F; Campbell, Matthew P; Cummings, Richard D; Feizi, Ten; Martin, Maria; Natale, Darren A; Packer, Nicolle H; Woods, Robert J; Agarwal, Gaurav; Arpinar, Sena; Bhat, Sanath; Blake, Judith; Castro, Leyla Jael Garcia; Fochtman, Brian; Gildersleeve, Jeffrey; Goldman, Radoslav; Holmes, Xavier; Jain, Vinamra; Kulkarni, Sujeet; Mahadik, Rupali; Mehta, Akul; Mousavi, Reza; Nakarakommula, Sandeep; Navelkar, Rahi; Pattabiraman, Nagarajan; Pierce, Michael J; Ross, Karen; Vasudev, Preethi; Vora, Jeet; Williamson, Tatiana; Zhang, Wenjin View Less	2020			iCite 7.83
GlyGen data model and processing workflow.					

Journal (Link to PubMed abstract)	Authors	Publication Year	Similar Publications	CitedBy	iCite RCR
Bioinformatics (Oxford, England) 2020 06 01; 36 (12) 3941-3943	Kahsay, Robel; Vora, Jeet; Navelkar, Rahi; Mousavi, Reza; Fochtman, Brian C; Holmes, Xavier; Pattabiraman, Nagarajan; Ranzinger, Rene; Mahadik, Rupali; Williamson, Tatiana; Kulkarni, Sujeet; Agarwal, Gaurav; Martin, Maria; Vasudev, Preethi; Garcia, Leyla; Edwards, Nathan; Zhang, Wenjin; Natale, Darren A; Ross, Karen; Aoki-Kinoshita, Kiyoko F; Campbell, Matthew P; York, William S; Mazumder, Raja View Less	2020			iCite 1.25

Virus-Receptor Interactions of Glycosylated SARS-CoV-2 Spike and Human ACE2 Receptor.

Cell host & microbe 2020 10 07; 28 (4) 586-601.e6	Zhao, Peng; Praissman, Jeremy L; Grant, Oliver C; Cai, Yongfei; Xiao, Tianshu; Rosenbalm, Katelyn E; Aoki, Kazuhiro; Kellman, Benjamin P; Bridger, Robert; Barouch, Dan H; Brindley, Melinda A; Lewis, Nathan E; Tiemeyer, Michael; Chen, Bing; Woods, Robert J; Wells, Lance View Less	2020			iCite 20.47

The GlySpace Alliance: toward a collaborative global glycoinformatics community.

Glycobiology 2020 01 28; 30 (2) 70-71	Aoki-Kinoshita, Kiyoko F; Lisacek, Frederique; Mazumder, Raja; York, William S; Packer, Nicolle H	2020			iCite 2.65

Tools for generating and analyzing glycan microarray data.

Beilstein journal of organic chemistry 2020; 16 2260-2271	Mehta, Akul Y; Heimburg-Molinaro, Jamie; Cummings, Richard D	2020			iCite 0.90

Glycomics-informed glycoproteomic analysis of site-specific glycosylation for SARS-CoV-2 spike protein.

STAR protocols 2020 12 18; 1 (3) 100214	Rosenbalm, Katelyn E; Tiemeyer, Michael; Wells, Lance; Aoki, Kazuhiro; Zhao, Peng	2020			iCite 1.41

GlyMDB: Glycan Microarray Database and analysis toolset.

Bioinformatics (Oxford, England) 2020 04 15; 36 (8) 2438-2442	Cao, Yiwei; Park, Sang-Jun; Mehta, Akul Y; Cummings, Richard D; Im, Wonpil	2020			iCite 1.08

Virus-Receptor Interactions of Glycosylated SARS-CoV-2 Spike and Human ACE2 Receptor.

bioRxiv : the preprint server for biology 2020 Jul 24;	Zhao, Peng; Praissman, Jeremy L; Grant, Oliver C; Cai, Yongfei; Xiao, Tianshu; Rosenbalm, Katelyn E; Aoki, Kazuhiro; Kellman, Benjamin P; Bridger, Robert; Barouch, Dan H; Brindley, Melinda A; Lewis, Nathan E; Tiemeyer, Michael; Chen, Bing; Woods, Robert J; Wells, Lance View Less	2020			iCite 0.83

GlycoGlyph: a glycan visualizing, drawing and naming application.

Bioinformatics (Oxford, England) 2020 06 01; 36 (11) 3613-3614	Mehta, Akul Y; Cummings, Richard D	2020			iCite 2.41

GLAD: GLycan Array Dashboard, a visual analytics tool for glycan microarrays.

Bioinformatics (Oxford, England) 2019 09 15; 35 (18) 3536-3537	Mehta, Akul Y; Cummings, Richard D	2019			iCite 2.01

Meeting Report of the International Life Science Integration Workshop 2018.

Journal (Link to PubMed abstract)	Authors	Publication Year	Similar Publications	CitedBy	iCite RCR
Glycobiology 2018 Aug 01; 28 (8) 552-555		2018			

Patents

No Patents information available for 5U01GM125267-04

Outcomes

The Project Outcomes shown here are displayed verbatim as submitted by the Principal Investigator (PI) for this award. Any opinions, findings, and conclusions or recommendations expressed are those of the PI and do not necessarily reflect the views of the National Institutes of Health. NIH has not endorsed the content below.

No Outcomes available for 5U01GM125267-04

Clinical Studies

No Clinical Studies information available for 5U01GM125267-04

News and More

Related News Releases

No news release information available for 5U01GM125267-04

History

No Historical information available for 5U01GM125267-04

Similar Projects

No Similar Projects information available for 5U01GM125267-04